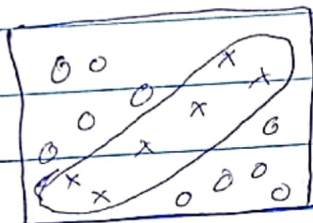


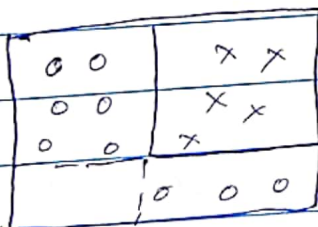
Section B & C

Q 1:

i) Not linearly separable, could use features such as $[1 \ x_1 \ x_2 \ x_1 x_2 \ x_1^2 \ x_2^2]$



ii) Not linearly separable, quadratic features could work $[1 \ x_1 > a \ x_2 > b]$



Q 2:

0	1	0	0	0
0	1	1	1	0
1	0	0	2	1
1	4	2	1	0
0	0	1	2	1

Q 3:

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{50}{51} = 0.98$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{50}{50+50} = 0.5$$

$$\text{FPR} = \frac{FP}{TN+FP} = \frac{1}{1+99} = 0.01$$

$$\text{TPR} / \text{FPR} = \frac{0.5}{0.01} = 50$$

$$\text{F-measure} = \frac{2 \times P \times R}{P+R} = \frac{2 \times 0.98 \times 0.5}{0.5+0.98} = 0.66$$

(Different variants of F-measure)